

Macrominerals and the Human Body



THE MACROMINERALS IN HUMAN MASS AND BIOLOGICAL FLUIDS

					BIOLOGICAL FLUID CONCENTRATION (MG·L ⁻¹)		
Macrominerals	Periodic Table Symbol	Ion Symbol in the Human Body	Atomic Weight	Total Amount in Human Body (g)	Plasma*	Sweat	Urine
Calcium	Ca	Ca ²⁺	40	900-1,500	85-105	0-40	100-180
Phosphorus	P	PO ₄ ³⁻	31	500-850	3,400-3,900	700-2,100	5,000-7,500
Potassium	K	K ⁺	39	130-180	16-30	4-15	60-100
Chloride	Cl	Cl ⁻	35.5	75-105	130-220	160-320	800-3,200
Sodium	Na	Na ⁺	23	65-105	20-50	3-6	20-1,100
Magnesium	Mg	Mg ²⁺	24	21-28	3,000-3,500	460-1,840	2,500-5,000

*Plasma and interstitial concentrations of ion are similar. Adapted from Gropper & Smith (2022) and Jeukendrup & Gleeson (2018).

MACROMINERALS AND STANDARD RECOMMENDATIONS

	RDA	RDA	AI	AI	AI	RDA	RDA
Sex	Age	Calcium/Ca ²⁺ (mg·d ⁻¹)	Phosphorus/PO ₄ ³⁻ (mg·d ⁻¹)	Potassium/K ⁺ (mg·d ⁻¹)	Chloride/Cl ⁻ (mg·d ⁻¹)	Sodium/Na ⁺ (mg·d ⁻¹)	Magnesium/Mg ²⁺ (mg·d ⁻¹)
Males	14-18	1,300	1,250	4,700	2,300	1,500	410
	19-30	1,000	700	4,700	2,300	1,500	400
	31-50	1,000	700	4,700	2,300	1,500	420
	51-70	1,000	700	4,700	2,300	1,300	420
	70+	1,200	700	4,700	1,800	1,200	420
Females	14-18	1,300	1,250	4,700	2,300	1,500	360
	19-30	1,000	700	4,700	2,300	1,500	310
	31-50	1,000	700	4,700	2,300	1,500	320
	51-70	1,200	700	4,700	2,300	1,300	320
	70+	1,200	700	4,700	1,800	1,200	320

Recommended Dietary Allowance (RDA); Adequate intake (AI); milligrams per day (mg·d⁻¹); micrograms per day (µg·d⁻¹). The table was adapted from (Gropper & Smith, 2022) for age categories in physique competitions.

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MACROMINERAL ABSORPTION, FUNCTIONS, AND SYMPTOMS RELATED TO DEFICIENCY OR TOXICITY

	Calcium/Ca ²⁺	Phosphorus/ PO ₄ ³⁻	Potassium/K ⁺	Chloride/Cl ⁻	Sodium/Na ⁺	Magnesium/Mg ²⁺
Mineral Required for Select Enzymatic Actions	Adenylate cyclase Ca/Mg-ATPase Calcineurin Glycerol-3-phosphate dehydrogenase Glycogen synthase Myosin kinase Pyruvate dehydrogenase	–	Pyruvate kinase	–	–	Required for >300 enzymatic reactions Glycolysis: Hexokinase, Glucokinase, and Phosphofructokinase PCr: Creatine kinase β-Oxidation: Acyl-CoA synthetase
Absorption in GI Tract (%)*	25-40	80-90	90-99	90-99	90-99	25-75
Cellular or Physiological Role	Component of bones and teeth; role in cellular processes, muscle contraction, membrane potentials, nerve impulse transmission, blood clotting, and enzyme activation.	Component of bone formation, buffer for muscle contraction, phospholipids, signaling compounds, nucleic acids, nucleotides, ATP-ADP phosphate transfer; pH regulation, cell membranes	Electrolyte, and pH (acid/base) balances; cell membrane polarization, nerve impulse generation, muscle contraction	Fluid and pH balance, component of gastric hydrochloric acid, chloride shift in RBCs	Blood volume homeostasis, pH (acid/base) balance, and electrolyte regulation; nerve transmission, muscle contraction	Component of bones, nerve impulse transmission, enzymatic reactions (metalloenzyme), ATPases, calcium regulation, ion transfer, promotes protein synthesis,
Deficiency	Rickets, osteoporosis, tetany	Neuromuscular, skeletal, hematologic, and cardiac symptoms; rickets, osteoporosis, muscle weakness, muscle cramps	Hypokalemia, muscular weakness, muscle cramps, cardiac arrhythmias, paralysis, apathy, loss of appetite	Convulsions (very rare excessive vomiting)	Hyponatremia, dizziness, coma muscle cramps, nausea, vomiting, loss of appetite, seizures, weight loss	Muscle weakness, fatigue, apathy, muscle tremor, cramps, neuromuscular hyperexcitability, cardiovascular effects and CNS effects
Excess or Toxicity	Impaired trace metal absorption, cardiac arrhythmia, constipation, kidney stones, calcification of soft tissue	Competitive of reduced absorption of Fe, Zn, and Cu; Ca ²⁺ metabolism compromised	Hyperkalemia, cardiac arrhythmia, cardiac failure	Hypertension (when consumed in excess with Na ²⁺), nausea	Hypertension, nausea	Nausea, vomiting

*The percentage of macromineral that is absorbed through the gut lining. More notable enzymes were selected for each mineral where appropriate. The unabsorbed portion will be excreted with feces. Iron (Fe), Zinc (Zn), Copper (Cu), Calcium (Ca²⁺), Na²⁺ (sodium ion); Adenosine triphosphate (ATP), Adenosine diphosphate (ADP), red blood cells (RBCs), Phosphocreatine (PCr), and central nervous system (CNS). Adapted from (Gropper & Smith (2022) and Jeukendrup & Gleeson (2018).

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TOLERABLE UPPER INTAKE LEVEL (UL) FOR MACROMINERALS

Sex	Age	Calcium/ Ca^{2+} ($\text{mg}\cdot\text{d}^{-1}$)	Phosphorus/ PO_4^{3-} ($\text{mg}\cdot\text{d}^{-1}$)	Potassium/ K^+ ($\text{mg}\cdot\text{d}^{-1}$)	Chloride/ Cl^- ($\text{mg}\cdot\text{d}^{-1}$)	Sodium/ Na^+ ($\text{mg}\cdot\text{d}^{-1}$)	Magnesium/ Mg^{2+} ($\text{mg}\cdot\text{d}^{-1}$)
Males and Females	14-18	3,000	4,000	5,000–7,000	3,600	2,300	350
	19-50	2,500	4,000	5,000–7,000	3,600	2,300	350
	51-70	2,000	4,000	5,000–7,000	3,600	2,300	350
	70+	2,000	3,000	5,000–7,000	3,600	2,300	350

*No tolerable upper intake level (UL) has been set for potassium due to insufficient data. Supplemental potassium doses of 5,000 to 7,000 $\text{mg}\cdot\text{d}^{-1}$ showed to cause an adverse effect on heart function in healthy adult subjects (Cappuccio et al., 2016). Adapted from Gropper & Smith (2022) for age categories in physique competitions.